

Mohammed Alshehri

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EDUCATION

University College Dublin

BSc Computer Science with Data Science

Dublin, IE

2024

Summary

AI/ML Research Engineer: **RL**, speech, multimodal systems, and **RL Ops**. Built and deployed Arabic-first LLM/TTS pipelines on **scalable cloud infrastructure**. Proven full-stack ML (React/Node) and **platform engineering** leadership at IBM/Taqirry.ai, improving quality/latency.

TECHNICAL SKILLS

ML/AI: PyTorch, Hugging Face Transformers, TensorFlow, **Reinforcement Learning (RL) Algorithms**, **RLlib**, LLM fine-tuning (SFT, LoRA, **RLHF principles**), evaluation, prompt engineering, ASR pipelines

Data/Infra: Python, Docker, Linux, REST APIs, Nginx, DigitalOcean, CI/CD basics

Backend: Node.js, Express.js

Frontend: React, JavaScript, HTML/CSS

EXPERIENCE

[redec.io](#) – Founding Lead Researcher & Engineer

10/2025 – Present

- **Built a voice-enabled patient education agent** for ReDec using foundation models to simplify lab analysis, post-clinical guidance, and patient follow-ups across speech and text; currently piloting with 2 clinics handling 150+ patient interactions weekly
- **Applied Reinforcement Learning** to fine-tune the conversational policy and implemented an evaluation framework essential for RLHF/Preference Tuning.

[Lisan.ai](#) – Research Engineer (Contract)

08/2025 – 01/2026

- **Designed and implemented an Arabic TTS** adaptation pipeline, incorporating RL DiffRO (Differentiable Reward Optimization) for emotional expressiveness and accent fine-tuning.
- **Built a continuous post-training** loop with automated eval gates and embedding-based clustering to expose failure modes and guide retraining.
- **Fine-tuned Arabic-first LLMs on domain datasets** (instruction tuning + preference tuning where applicable); significantly improving task accuracy on internal eval sets facilitating subsequent deployment.

[Taqirry.ai](#) – Founding Lead Research

12/2024 – 08/2025

- **Co-founded and led ML + product engineering for an AI notetaker**, optimizing open-source models into production-ready transcription, summarization, and automation.
- **Built an Arabic ASR + reasoning system with** applied RAG, optimizing LLM inference (caching, chunking, streaming) for an on-prem deployment; improved accuracy and cut end-to-end latency by 40%.
- **Actively engaged in sales** (demos, client objections), closing deals, including with B2G governmental entities.

IBM – Multimodal AI Engineer

06/2024 – 12/2024

- **Built and optimized multimodal AI systems** combining ASR and LLM prompt engineering to improve speech and text understanding in enterprise AI products.
- **Contributed to the Watsonx team**, improving semantic accuracy and human–AI interaction across voice and text pipelines.
- **Designed evaluation frameworks**, ran model validation, and collaborated on applied experiments utilizing OpenAI API endpoints to benchmark against internal systems

IBM – NLP, AI, Software Engineer Intern

03/2023 – 09/2023

- **Engineered an advanced chatbot utilizing large language models** which simplified legal document interactions, achieving a 30% reduction in customer query handling time.
- **Enhanced the chatbot's architecture** to deliver 25% quicker and more precise human-like responses, significantly boosting real-time legal jargon interpretation using NLP techniques.